

From "Steel Slag Dump" to "Ecological Mine-Park": A Study on Multi-Waste Synergistic Transformation and Carbon Sequestration Integration Based on Accelerated Artificial Diagenesis

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Abstract

China's iron and steel industry produces over 100 million tonnes of steel slag annually, with historical cumulative stockpiles exceeding 1.5 billion tonnes and a comprehensive utilization rate below 30%. The implementation of GB 175-2023 "Common Portland Cement" has further narrowed the direct disposal pathways for steel slag in cement-based materials. Departing from the conventional engineering mindset of "rapid disposal," this paper proposes a paradigm shift across spatial and temporal scales—redefining steel slag landfills as an "artificial vein-ore" ecosystem, in which steel slag serves as the inorganic skeleton, organic waste as the biochemical engine, and functional microorganisms together with saprophagous insects act as synergistic drivers. For the first time, we systematically propose the theory of "Accelerated Artificial Diagenesis (AAD)." This theory compresses geological-scale natural diagenesis into an engineering scale of approximately five years through multi-factor synergistic enhancement, achieving the fundamental transformation of industrial solid waste into stable "artificial ore" via a three-stage ecological succession of "activation—stabilization—mineralization." The paper systematically elucidates the multi-waste synergistic symbiosis mechanism, the microbial mineralization-accelerated carbonation

mechanism, the black soldier fly bioconversion mechanism, the mechanism of active utilization of steel slag spontaneous pulverization, and key biochemical coupling processes including organic acid–calcium ion release coupling, bioheat generation–self-heating maintenance coupling, in-situ CO₂ supply–carbonation coupling, and straw skeleton–permeability enhancement coupling. Based on the demonstration of technical feasibility, this paper presents a complete engineering design for a 10,000-tonne-scale proof-of-concept project, together with a full life-cycle carbon footprint analysis and a preliminary economic feasibility assessment. The core insight of this study is as follows: acknowledging the inherent limitations of engineering intervention, we turn to learning from nature, harnessing time and biological communities—the two most powerful forces at our disposal—to transform today's "environmental liabilities" into tomorrow's "resource assets," thereby exploring a novel paradigm for industrial civilization to transition toward a "geological-scale circular economy" in harmony with the Earth system.

Keywords: steel slag valorization; accelerated artificial diagenesis; microbially induced calcium carbonate precipitation (MICP); multi-waste synergy; black soldier fly; carbon sequestration; artificial vein-ore; ecological mine-park

1 Introduction

1.1 Scale and Urgency of the Steel Slag Problem

China is the world's largest crude steel producer, with an annual crude steel output consistently maintained at the order of 1 billion tonnes, accounting for over 53% of global production (National Bureau of Statistics, 2024). Based on a generation rate of 100–150 kg of steel slag per tonne of crude steel, China's annual steel slag production amounts to approximately 100–150 million tonnes. The mineralogical composition of steel slag is dominated by dicalcium silicate (C₂S), tricalcium silicate (C₃S), calcium ferrite (C₂F), and the RO phase (MgO-FeO-MnO solid solution). Its chemical composition (CaO 40–50%, SiO₂ 10–15%, Al₂O₃ 1–5%, FeO/Fe₂O₃ 10–30%, MgO 5–10%) bears a certain resemblance to that of cement clinker, theoretically offering potential as a supplementary cementitious material.

However, steel slag contains 2–10% free calcium oxide (f-CaO) and a certain amount of free magnesium oxide (f-MgO). The hydration reaction of f-CaO proceeds at an extremely slow rate, and the formation of $\text{Ca}(\text{OH})_2$ through this hydration is accompanied by approximately 97% volume expansion, a process that can persist for decades or even centuries under natural conditions. When steel slag containing insufficiently slaked f-CaO is used as a building material, the risk of cracking due to delayed expansion poses a critical unsoundness problem. China's comprehensive utilization rate of steel slag has long remained below 30%, with historical cumulative stockpiles exceeding 1.5 billion tonnes (Zhang, 2026).

Applicability analysis of different types of steel slag. The mineralogical composition and f-CaO content of steel slag vary significantly depending on the steelmaking process, which directly affects the transformation cycle and process parameters within the "Jianggang Plan." Basic oxygen furnace (BOF) slag typically contains 3–8% f-CaO and a relatively high FeO content (15–25%), with a mineralogical assemblage dominated by C_2S , C_3S , C_2F , and the RO phase. BOF slag is an ideal feedstock for the "Jianggang Plan," as its moderate f-CaO content renders transformation within a five-year cycle highly feasible. Electric arc furnace (EAF) oxidizing slag has an f-CaO content comparable to that of BOF slag (3–8%) but a higher FeO content (20–35%) and greater density, requiring consideration of its high density during stockpile construction. EAF reducing slag can have an f-CaO content as high as 15% or more, along with a substantial proportion of C_3A (tricalcium aluminate), resulting in extremely high reactivity. For this type, it is recommended to extend the transformation cycle to 7–8 years, appropriately reduce the steel slag layer thickness (to 10–15 cm), increase the organic acid contact area, or adopt a layered alternating arrangement with other types of steel slag. Ladle furnace slag has an extremely high CaO content (50–60%), a high Al_2O_3 content (20–30%), and an f-CaO content reaching 10–20%. It is recommended to blend it with BOF slag at a proportion not exceeding 30%. Stainless steel slag has a high Cr content (2–5%), and special attention must be paid to the leaching risk of Cr(VI). Its inclusion in the routine material system of the "Jianggang Plan" is not recommended at present, pending the maturation of heavy metal stabilization technologies.

Synergistic effects of mixed steel slags. The blended system of high-calcium ladle furnace slag (CaO 50–60%) and BOF slag (CaO 40–45%) exhibits significant complementarity: the ladle furnace slag provides an abundant supply of Ca^{2+} , while the BOF slag supplies Si^{4+} and Al^{3+} , forming a $\text{CaO-SiO}_2\text{-Al}_2\text{O}_3$ system that more closely approximates the chemical composition of cement clinker and favors the formation of composite minerals such as calcium carboaluminate during the carbonation stage. Blending EAF reducing slag (high f-CaO) with BOF slag (moderate f-CaO) can effectively "dilute" the f-CaO concentration in the system, preventing temperature runaway and structural damage caused by excessively localized vigorous reactions. It is recommended that the proportion of high-f-CaO slag in the blended system not exceed 30% (by mass).

Effect of steel slag cooling method on reactivity. The cooling method employed for steel slag exerts a decisive influence on its mineralogical composition and reactivity. Water-quenched steel slag (rapid cooling), owing to its extremely high cooling rate ($>100^\circ\text{C/s}$), prevents the crystallization of molten silicates, resulting in the formation of a substantial vitreous phase content (up to 30–50%). The Ca^{2+} and Si^{4+} sequestered within the vitreous phase exist in a high-energy state, conferring strong reactivity, while f-CaO is predominantly present in microcrystalline or amorphous form and dissolves more rapidly under organic acid conditions. Air-quenched steel slag (intermediate cooling) has a vitreous phase content (approximately 15–30%) that lies between that of water-quenched and naturally cooled slags, with a moderate degree of mineral crystallinity. Naturally cooled steel slag (slow cooling) exhibits high crystallinity, with f-CaO present as relatively large crystalline grains (up to tens of micrometers) often encased by minerals such as C_2S and C_3S , resulting in low reactivity. For the "Jianggang Plan," water-quenched steel slag is the most suitable primary feedstock, as its high vitreous phase content and finely dispersed f-CaO facilitate a shorter transformation cycle. Naturally cooled steel slag requires an appropriate extension of the activation time in Stage I (from 2 years to 3 years), or compensation for its low reactivity by increasing the organic acid supply.

1.2 Policy Environment: Structural Transition from Acquiescence to Tightening

The national standard GB 175-2023 "Common Portland Cement," released in 2023, imposes stricter limits on the types and proportions of admixtures permitted in cement, and steel slag has been explicitly excluded from the list of allowable admixtures for common Portland cement. This regulatory adjustment has effectively closed the traditional primary pathway for steel slag disposal, rendering the issue of steel slag valorization even more pressing.

Concurrently, the "Comprehensive Solid Waste Treatment Action Plan" (Guo Fa [2025] No. 14, referred to in the industry as the "Ten Measures for Solid Waste"), issued by the State Council in December 2025, explicitly sets binding targets by 2030, including: achieving a national annual comprehensive utilization volume of 4.5 billion tonnes for bulk industrial solid waste, effective control of historical stockpiles, and a substantial increase in the comprehensive utilization rate of newly generated smelting slag. This combination of policy "tightening on one front and opening on another" provides an unprecedented strategic demand and policy window for fundamental innovation in steel slag valorization technologies.

1.3 Limitations Analysis of Mainstream Technical Pathways

Current steel slag treatment technologies can be categorized into three major routes: physical digestion, chemical carbonation, and surface modification.

The physical digestion route includes the hot-stewing method, the autoclave method, and the natural weathering method. The hot-stewing method utilizes the residual heat of steel slag by spraying water to generate steam, which accelerates the hydration of f-CaO under high temperature and high pressure, reducing the f-CaO content to below 3%. However, the penetration depth of high-pressure steam is limited, making it difficult to thoroughly digest f-CaO inside large slag blocks, and both energy consumption and equipment investment costs are relatively high. The natural weathering method requires

open-air stockpiling of steel slag for 1-3 years; although the cost is low, it occupies extensive land area and the completeness of digestion is difficult to guarantee.

The chemical carbonation route utilizes CO_2 to react with f-CaO to form stable CaCO_3 . However, under atmospheric pressure, the carbonation rate of f-CaO is severely limited by CO_2 mass transfer and the encapsulation effect of the product layer. Although pressurized carbonation can accelerate the reaction, the equipment investment and operating costs increase significantly.

The surface modification route wraps steel slag particles with organic/inorganic coatings to block moisture contact. Its effect is immediate, but the long-term reliability of coating durability over decades of building service life has yet to be fully validated.

The common limitation of the above technical routes lies in the attempt to "conquer" an intrinsically geological-scale chemical reaction process within an extremely short engineering timeframe using high energy input, resulting in varying degrees of a "cost-effectiveness-scale" impossible triangle.

1.4 Paradigm Shift and Core Proposition of This Study

This paper proposes a fundamental epistemological shift: acknowledging the limitations of engineering means and transforming the logic from "rapid treatment" to "controlled transformation." Inspired by the wisdom of the traditional Chinese "Jianggang" (fermentation vat) long-term fermentation and the natural diagenetic processes in geology, this paper systematically proposes, for the first time, the theoretical framework of "Accelerated Artificial Diagenesis" (AAD).

The core proposition of this theory is: redefining the steel slag landfill site as a human-designed and controlled giant surface weathering-sedimentation reactor. With a complete cycle of 5 years, through the synergistic coexistence of multi-source wastes and the introduction of functional biological communities, the thorough mineralization and transformation of unstable components in steel slag is systematically completed under controlled conditions. Over time, the stockpile site successively evolves into a "natural

ecological park" and an "artificial ore park," ultimately achieving a complete transformation from an "environmental burden" to a "resource reserve."

2 Theoretical Framework: Scientific Basis and System Design of the Accelerated Artificial Diagenesis Theory

2.1 Accelerated Artificial Diagenesis Theory

In geology, "diagenesis" refers to all physical, chemical, and biological changes that sediments undergo after deposition and before metamorphism, ultimately transforming loose sediments into consolidated sedimentary rocks. The key processes of diagenesis include compaction and dewatering, dissolution and reprecipitation, mineral phase transformation, microbial activity, and the formation of carbonate cements. The timescale of natural diagenesis is typically on the order of millions to tens of millions of years.

The Accelerated Artificial Diagenesis (AAD) theory proposed in this paper refers to the engineering intensification of the driving forces of diagenesis (water, CO₂, microorganisms) under human intervention, and the engineering design of its spatial (stockpile) and temporal (5-year cycle) dimensions, thereby compressing the diagenetic process that requires geological epochs in nature to an engineering-accessible timescale.

The core strategy of the AAD theory is "multi-factor synergistic intensification": simultaneously applying three types of driving forces within the stockpile — physical intensification (leachate recirculation, pulsed spraying, particle size gradation optimization), chemical intensification (organic acid-promoted f-CaO dissolution, CO₂ injection), and biological intensification (inoculation with functional microbial consortia and insects) — to generate synergistic amplification effects, with the overall transformation efficiency far exceeding the simple superposition of individual approaches.

Systematic comparison between artificial diagenesis and natural diagenesis. To clearly reveal the essential characteristics of the AAD theory, Table 1 provides a systematic comparison across dimensions including driving forces, timescale, temperature conditions, CO₂ supply, microbial action, diagenetic products, and product homogeneity.

Table 1 Systematic comparison between natural diagenesis and accelerated artificial diagenesis (AAD)

This table clearly reveals the engineering essence of AAD: through multi-factor synergistic intensification, the driving forces of natural diagenesis are multiplicatively enhanced, thereby achieving a leapfrog compression of the timescale. This theory constitutes the scientific core of the entire "Jianggang Plan," and all subsequent technical designs are developed based on this foundation.

2.2 Multi-Source Waste Synergistic Symbiosis System

The greatest innovation of the "Jianggang Plan" that distinguishes it from existing solid waste treatment technologies lies in breaking the industry convention of "classified treatment" and proactively constructing a synergistic symbiosis system of multi-source wastes.

2.2.1 Material Components and Functional Division

The main material components of the system and their core functions are shown in Table 2.

Table 2 Material components and functional division of the multi-source waste synergistic symbiosis system

2.2.2 Key Chemical-Biological Coupling Mechanisms of Multi-Source Synergy

The synergistic transformation of the above materials within the "Jianggang" depends on the following key coupling processes:

(a) Organic acid neutralization-calcium ion release coupling

Organic wastes produce large amounts of organic acids (acetic acid, propionic acid, butyric acid, lactic acid, etc.) during the microbial anaerobic hydrolysis-acidification

stage, reducing the local environmental pH to 4-6. The reaction of these organic acids with f-CaO in steel slag is shown in Equation (1):

(1)

This reaction has three synergistic benefits: the organic acid "in-situ etches" the inert hydration layer on the f-CaO surface, exposing fresh reaction interfaces; the generated calcium acetate has extremely high solubility (approximately 34.7 g/100 mL H₂O, 20°C), effectively activating Ca²⁺ into the solution phase; and the Ca²⁺ in the solution phase is in a highly reactive state, readily combining with CO₃²⁻ in the system to form CaCO₃ precipitates, thereby completing the full "dissolution-migration-mineralization" transformation chain.

(b) Active utilization mechanism of steel slag self-pulverization

The hydration expansion of f-CaO in steel slag is often considered a detrimental phenomenon, but in the AAD system, it can be actively converted into an advantage. When moisture penetrates into the interior of steel slag blocks through pores, the localized expansion stress (up to tens of MPa) generated by the hydration of f-CaO to form Ca(OH)₂ induces the self-pulverization of steel slag — large slag blocks disintegrate into fine particles (particle size can decrease from tens of centimeters to a few millimeters). This process greatly increases the specific surface area of the steel slag (by 2-3 orders of magnitude), significantly enhancing the mass transfer efficiency of subsequent organic acid dissolution and CO₂ carbonation reactions. In the early stage of the activation period, through reasonable water spraying scheduling (under stockpile temperature conditions of 50-60°C), this transformation from "detrimental expansion" to "beneficial pulverization" can be initiated and accelerated.

(c) Biological heat generation-self-heating maintenance coupling

Organic wastes release large amounts of metabolic heat during aerobic degradation. In a composting system, the internal temperature during the mesophilic-thermophilic phase can reach 50-75°C. According to the Arrhenius equation, the chemical reaction rate constant is exponentially related to temperature ($k = A \cdot e^{-E_a/RT}$). The self-heating of the

stockpile raises the reaction temperature from ambient temperature (approximately 20°C) to 55-65°C, which can increase the f-CaO carbonation rate by 5-10 times and the hydration reaction rate by 10-20 times. This "self-heating-driven" effect enables a substantial increase in reaction rates within the system without the need for external energy input.

(d) In-situ CO₂ supply-carbonation coupling

During the degradation of organic wastes, aerobic respiration produces CO₂ (approximately 0.5-1.0 kg CO₂/kg organic matter dry weight), and the acidogenic phase prior to anaerobic methanogenesis also releases large amounts of CO₂. Based on the release of approximately 500 kg CO₂ per ton of organic waste, a ten-thousand-ton stockpile can generate several thousand tons of CO₂ annually. This portion of CO₂ is generated in situ within the stockpile and undergoes carbonation reactions upon contacting Ca²⁺ in the solution phase through pore diffusion (Equation 2):

(2)

The greatest advantage of in-situ CO₂ supply is that the reaction occurs in the micro-to-millimeter-scale pore space within the stockpile, overcoming the mass transfer bottleneck of CO₂ diffusion from the bulk gas phase to the interior of solid particles that plagues external pressurized carbonation processes.

(e) Straw skeleton-ventilation enhancement coupling

The fibrous porous structure of straw (cellulose content approximately 35-45%, hemicellulose approximately 25-35%, lignin approximately 15-20%) makes it a recalcitrant organic material. In the stockpile, the physical structure of straw creates interconnected macro-pore networks (pore diameter 0.1-10 mm), providing efficient mass transfer channels for the inward diffusion of O₂ and the outward migration of CO₂, thereby avoiding the issues of hindered gas diffusion and localized anaerobic dead zones that arise from the tight packing of fine particles in pure steel slag stockpiles.

2.2.3 Microscopic Mechanisms of Microbe-Mineral Interfacial Interactions

Microbial transformation of steel slag minerals does not occur throughout the entire solution but is highly concentrated at the mineral-biofilm interface. Microorganisms colonize the mineral surface and create localized reaction microenvironments through the following mechanisms: secreting extracellular polymeric substances (EPS) to form biofilms, constructing a reaction layer approximately 10-100 μm thick on the mineral surface; within the biofilm, the concentration of organic acids produced by microbial metabolism can be 1-3 orders of magnitude higher than that in the bulk solution, forming a microzone acidic environment that efficiently drives mineral dissolution; microorganisms actively take up essential elements such as Ca^{2+} and Mg^{2+} from the mineral surface through active transport, and induce mineral precipitation on the cell wall or EPS template through metabolic activity.

The important implication of this microscopic mechanism for stockpile design is that microorganisms need full contact with the steel slag mineral surface to achieve maximum transformation efficiency. Therefore, the bacterial suspension should uniformly penetrate various parts of the steel slag layer, rather than remaining solely in the pore solution of the stockpile.

2.3 Biological Synergy: Microbial Mineralization and Insect Bioconversion

The core design of the Jianggang (fermentation vat) Plan at the biological community level is the establishment of a three-tier synergistic transformation system comprising "microbes—insects—plants."

2.3.1 Functional Microbial Communities

Microorganisms serve as the "biochemical engine" of the AAD system, with their core functions encompassing organic matter degradation, organic acid production, and

calcium carbonate mineralization. The functional microbial community designed in this concept includes the following key groups:

Hydrolytic acidogenic bacteria (anaerobic/facultative anaerobic): Responsible for decomposing complex organic matter such as cellulose, proteins, and lipids into small-molecule organic acids, amino acids, and monosaccharides. This group primarily includes cellulose-degrading bacteria (*Clostridium* spp., *Ruminococcus* spp.) and acidogenic bacteria (*Lactobacillus* spp., *Propionibacterium* spp.). These bacterial groups have been widely applied in natural composting and biogas fermentation systems; upon inoculation, they can initiate the acidification process of the pile body within 2–3 days.

Urea-hydrolyzing bacteria (facultative): Catalyze urea hydrolysis via urease secretion to produce CO_3^{2-} and NH_4^+ (Eq. 3), promoting CaCO_3 precipitation in an alkaline microenvironment.

(3)

Sporosarcina pasteurii is the flagship strain in MICP technology, exhibiting urease activity of 1–10 mmol urea/min/mg protein and capable of inducing significant CaCO_3 precipitation within hours to days. Previous studies have demonstrated that this strain can form a dense calcium carbonate shell on the surface of steel slag, effectively encapsulating f-CaO particles and preventing their subsequent hydration.

Carbonic anhydrase (CA)-producing bacteria: Catalyze the CO_2 hydration reaction efficiently via CA enzyme secretion (Eq. 4), increasing the conversion rate by 10^4 – 10^6 fold.

(4)

Research by Professor Qian Chunxiang's team has demonstrated that suspended CA-producing microorganisms can reduce the time required to achieve 10% carbon sequestration in steel slag to one-fifth of that needed by direct carbonation, revealing the essence of microbial enhancement of carbonation kinetics through multiple mechanisms including promoting CO_2 dissociation and diffusion within the liquid film and accelerating the conversion of silicate minerals such as C_2S .

Sulfate-reducing bacteria (SRB): Can utilize sulfate-containing minerals such as gypsum to generate S^{2-} , inducing the precipitation of stable minerals such as vivianite ($Fe_3(PO_4)_2 \cdot 8H_2O$) in phosphorus-rich environments, while simultaneously releasing OH^- to elevate pH and initiate carbonate mineralization. When the phosphorus content of steel slag is relatively high, SRB can serve as an important supplementary functional bacterial group.

Microbial community succession and functional redundancy. The microbial community within the pile body will undergo a predictable ecological succession: Stage 1 (activation period), characterized by abundant organic matter and large pH fluctuations, during which fast-growing r-strategists (lactic acid bacteria, yeasts, etc.) dominate; Stage 2 (stabilization period), during which organic matter is progressively consumed, pH stabilizes at 8–10, and slow-growing K-strategists (mineralizing bacteria such as *Sporosarcina pasteurii*) establish dominance; Stage 3 (mineralization period), characterized by nutrient depletion, during which oligotrophic bacteria and endospore-forming bacteria become the dominant community members. To ensure functional redundancy, a multi-strain mixed inoculation strategy is adopted—urease-producing bacteria can be simultaneously inoculated with *Sporosarcina pasteurii*, *Bacillus sphaericus*, and *Bacillus megaterium*, such that if one strain declines, the others can substitute for its function.

Microbial community monitoring and early warning. It is recommended to establish a microbial community monitoring and early warning system based on 16S rRNA high-throughput sequencing: samples should be collected from different depths of the pile body every 3 months to analyze changes in community composition and diversity. When the relative abundance of functional bacterial groups declines by more than 50% from the peak value, or when the Shannon diversity index decreases by more than 30% from the baseline, an early warning should be triggered and supplementary inoculation procedures initiated.

2.3.2 Black Soldier Fly Bioconversion System

The black soldier fly (*Hermetia illucens* L.), belonging to the family Stratiomyidae of the order Diptera, has emerged in recent years as the most widely studied saprophagous insect species in the field of organic solid waste bioconversion. Its larvae feed on organic waste and possess outstanding advantages including a broad dietary range, high conversion efficiency, rapid growth rate, and non-vector status for disease transmission.

Core conversion capacity. Under conditions of 25–30°C, black soldier fly larvae require approximately 25 days from hatching to prepupal stage; one ton of larvae can process 4 tons of livestock manure or 2.5 tons of food waste within a rearing cycle of approximately 40 days. The larval body contains 35–45% crude protein (dry weight) and 25–35% crude fat (dry weight), while the frass is rich in humic acid and nitrogen, phosphorus, and potassium. The recommended optimal inoculation density is 5,000–10,000 individuals/m² (calculated based on the surface area of the organic mixture layer); below the lower limit, conversion is insufficient, while above the upper limit, intraspecific competition is triggered. In large-scale operations, a batch inoculation strategy can be adopted: initial inoculation at 5,000 individuals/m², followed by supplementation with 2,000–3,000 individuals/m² after 2 weeks depending on consumption status.

Synergistic bioconversion by black soldier fly and microorganisms. The larval gut harbors a rich microbial community (Proteobacteria, Bacteroidetes, Firmicutes, etc.), which is excreted into the pile substrate during feeding, assisting in the decomposition of recalcitrant organic matter such as cellulose, and secreting urease and carbonic anhydrase to indirectly promote CaCO₃ precipitation. The feeding and burrowing activities of larvae significantly alter the microbial community structure of the pile body: by ingesting organic matter, they pulverize large material fragments into fine particles, substantially increasing the surface area available to microorganisms (by 10–100 fold); the micro-channels created within the pile body improve oxygen diffusion and gas exchange; and the excreted frass, rich in readily degradable organic matter and active microorganisms, forms distributed mineralization "hotspots" within the pile.

Potential of other insect species. In addition to the black soldier fly, larvae of the housefly (*Musca domestica*) can convert livestock manure within 4–5 days, with a protein

content as high as 50–60%; the yellow mealworm (*Tenebrio molitor*) can feed on materials with higher cellulose content such as straw, and its gut microbiota possess unique lignin-degrading capabilities; larvae of the white-spotted flower chafer (*Protaetia brevitarsis*) can feed on mature compost and straw, producing high-quality frass rich in organic matter and trace elements. In the future, suitable combinations of insect species can be flexibly selected based on material characteristics and climatic conditions of different regions.

2.4 Five-Year Ecological Succession Model

The core temporal framework of the Jianggang Plan is a complete cycle of 5 years, during which the pile body undergoes three distinct ecological succession stages. Table 3 systematically aligns these ecological succession stages with the corresponding phases of geological diagenesis.

Table 3 Correspondence between the five-year ecological succession model of the Jianggang Plan and diagenetic stages

Porosity evolution patterns. In Stage 1, porosity is relatively high (35–45%), dominated by macropores formed by organic matter decomposition; in Stage 2, porosity gradually decreases (25–35%), as fine CaCO_3 crystals begin to precipitate within the pores; in Stage 3, porosity further declines (20–30%), as macropores are filled with CaCO_3 and C-S-H gel, and connected porosity decreases significantly—a key factor in the strength enhancement of "artificial ore." Calcite is expected to be the dominant cementation crystal phase (thermodynamically most stable), with C-S-H gel serving as a secondary cementitious phase filling intergranular pores. The compressive strength of ore dominated by calcite pore-filling cementation can reach 10–25 MPa.

3 Technical Feasibility Assessment

3.1 Research Progress on MICP-Accelerated Steel Slag Carbonation

A study by Professor Qian Chunxiang's team at Southeast University, published in *Engineering* (the journal of the Chinese Academy of Engineering), combined microbial

technology with a rotary process, using carbonic anhydrase-producing microorganisms to accelerate CO₂ fixation from cement kiln exhaust gas in steel slag powder, achieving a CO₂ fixation rate more than double that of the control. Further research by the same team employed suspended-state MICP technology, reducing the time required to achieve 10% carbon sequestration to one-fifth of that needed by direct carbonation, and elucidated the multiple acceleration mechanisms. When the carbon sequestration rate exceeds 8% and the specific surface area is no less than 300 m²/kg, the volume stability issue of steel slag can be effectively resolved (Qian et al., 2025).

Regarding direct improvement of volume stability, *Sporosarcina pasteurii* has been demonstrated to achieve synchronous CO₂ mineralization sequestration and f-CaO removal under ambient conditions (30°C, atmospheric pressure). A multi-microorganism (*Sporosarcina pasteurii* + yeast + carbonic anhydrase-producing bacteria) synergistic pre-treatment method for steel slag has also been proposed. A green modification route combining carbonic anhydrase-mediated microbial mineralization with boiling water pre-treatment has been demonstrated at both macro- and micro-scales to effectively improve the volume stability of steel slag fine aggregates.

3.2 Research on Synergistic Treatment of Steel Slag and Organic Waste

Previous studies have successfully used steel slag as the primary cementitious material, incorporating rice straw fibers to prepare foamed concrete, demonstrating the feasibility of synergistic resource utilization of agricultural and industrial by-products. A patented technology (CN120647217A) for preparing asphalt concrete using modified steel slag combined with iron tailings and corn straw has confirmed the technical pathway for synergistic utilization of effective components from multiple waste streams.

In the field of environmental remediation, composite materials of steel slag and rice straw biochar have exhibited excellent performance in the immobilization of heavy metals (e.g., arsenic) in soil. Field trials have demonstrated that combined application of steel slag and green waste compost significantly promotes the growth of durum wheat, improves soil health, and enhances drought resistance (2025).

At the industrial demonstration level, the CCUS technology industrial demonstration project at HBIS Group Tangsteel New Area achieved, for the first time at industrial scale, the integration of two technical routes: "blast furnace gas carbon capture coupled with microbial carbon fixation for protein production" and "steel slag flue gas carbonation for resource utilization," producing high-quality building materials with a carbon sequestration capacity of 20% and compressive strength exceeding 30 MPa, which received the "CarbXRock" trademark authorization (HBIS Group, 2026).

3.3 Large-Scale Black Soldier Fly Bioconversion Practice

In China, the industry-scale treatment of organic waste using the black soldier fly has become quite mature. The Yantian District of Shenzhen has established the nation's first full-scale black soldier fly food waste treatment demonstration zone, cumulatively processing over 55,000 tons of food waste, with an estimated carbon emission reduction of 326 tons of CO₂ equivalent in 2025. The research team led by Professor Huang Yongping at Shanghai Jiao Tong University has built a black soldier fly pilot line with a daily processing capacity of 4 tons of food waste on campus. A black soldier fly project in Fuyang, Hangzhou, with a daily processing capacity of 30 tons of organic waste, has obtained commercial financial institution loan support. These cases fully demonstrate the engineering feasibility and economic sustainability of black soldier fly technology in the field of organic waste treatment.

3.4 Remanufacturing Pathways for Artificial Ore

The end product of the AAD process—"artificial ore"—is dominated by CaCO₃ and C-S-H gel as its primary mineral phases and is completely stabilized in nature. Its utilization as building materials can follow four technical pathways depending on the target product:

Autoclaved products pathway. The artificial ore is crushed, mixed with appropriate amounts of quicklime and gypsum in a specified ratio, and reacted in an autoclave (approximately 180–200°C, approximately 1.0–1.5 MPa saturated steam pressure) for 6–10 hours to produce autoclaved lime-sand bricks or aerated concrete blocks with

tobermorite ($5\text{CaO}\cdot 6\text{SiO}_2\cdot 5\text{H}_2\text{O}$) as the main hydration product, ensuring fully guaranteed volume stability.

Cement clinker pathway. The chemical composition of artificial ore ($\text{CaO-SiO}_2\text{-Al}_2\text{O}_3\text{-Fe}_2\text{O}_3$) corresponds to a mixed mineral of "artificial limestone + clay." After crushing and homogenization, it is fed into a rotary kiln and calcined at $1300\text{--}1450^\circ\text{C}$, where the components recombine to form new cement clinker dominated by C_3S and C_2S . This pathway is essentially a "secondary calcination and remanufacturing" of steel slag after homogenization and stabilization in the AAD stage, thoroughly eliminating volume stability risks at the raw material level.

Carbonated building materials pathway. The artificial ore is crushed, pressed into shape, and cured in a CO_2 atmosphere (concentration $> 20\%$) for several hours to days. Residual C-S-H gel reacts with CO_2 to form CaCO_3 and silica gel ($\text{SiO}_2\cdot n\text{H}_2\text{O}$); the silica gel fills pores while CaCO_3 serves as a cementitious phase, forming a dense calcium carbonate-silica gel composite matrix. Steel mills' blast furnace gas (CO_2 content $20\text{--}25\%$) or cement plant kiln exhaust gas (CO_2 content $15\text{--}30\%$) can be used as CO_2 sources. The HBIS Group Tangsteel New Area CCUS project has already demonstrated the full industrial feasibility of this route.

Concrete aggregate pathway. After 5 years of mineralization stabilization, the f-CaO content of artificial ore drops below 1% , and its volume stability fully meets the requirements for concrete aggregates. Its apparent density is approximately $2.4\text{--}2.8\text{ g/cm}^3$, water absorption rate approximately $3\text{--}6\%$, and crushing index approximately $10\text{--}15\%$, meeting the Class II or Class III aggregate standards of GB/T 14685. Compared with natural crushed stone, artificial ore aggregates possess unique advantages including a rough porous surface, strong mechanical interlocking force with cement paste, a dense interfacial transition zone, and residual CaCO_3 reactivity that can continuously participate in carbonation-based micro-repair during the service life of concrete. With China's annual concrete production exceeding 3 billion cubic meters and natural sand and gravel aggregate resources becoming increasingly depleted, the potential market space for artificial ore as a substitute aggregate is enormous.

Artificial ore reactivity evaluation methods. A dedicated multi-index evaluation system needs to be established: calcium carbonate content (TGA method, 600–850°C weight loss attributed to CaCO_3 decomposition), residual f-CaO content (ethylene glycol-ethanol method, GB/T 176), soluble CaO content (ammonium acetate solution extraction), strength activity index (mixed with reference cement at a 3:7 ratio, 28-day compressive strength ratio), etc.

A techno-economic comparison of the four pathways is presented in Table 4.

Table 4 Techno-economic comparison of the four remanufacturing pathways for artificial ore

Standardization work plan: Years 1–2: complete the classification and nomenclature specification for "artificial ore" (group standard); Years 3–4: complete the product quality standard (upgrade from group standard to industry standard); Years 5–6: complete the application technical specifications for autoclaved products/carbonated building materials; Year 7 and beyond: initiate the national standard declaration process.

4 Preliminary Design of the Proof-of-Concept Engineering System

4.1 Site Selection and Scale

Site Selection Criteria: The site shall be established on existing steel slag yards or idle industrial lands associated with iron and steel enterprises, with priority given to the utilization of abandoned industrial sites and degraded land. The site should be adjacent to urban kitchen waste treatment plants and agricultural cultivation areas (to facilitate straw collection and transport). The total land area is approximately 2–3 ha (30–45 mu).

Treatment Capacity: Total processing volume is approximately 10,000 m³, of which steel slag accounts for about 60% (6,000 m³, ~12,000 tons) and organic waste (straw, kitchen waste, livestock manure, and biochar) accounts for approximately 40%. The annual treatment capacity is approximately 2,400 tons of steel slag and 1,600 tons of organic waste.

4.2 Heap Construction and Layer Sequence Design

A "sandwich" layered sequence structure is adopted, arranged from bottom to top as follows:

(a) Foundation layer (30 cm): A composite anti-seepage system comprising compacted clay + a double-layer HDPE geomembrane (2 mm) + a geosynthetic clay liner (GCL). An electrically conductive leak detection grid is installed between the two geomembrane layers; in the event of a leak, the resistance of the detection layer changes, enabling immediate localization of the leak point and initiation of emergency response measures.

(b) Drainage layer (20 cm): Composed of gravel and a network of corrugated perforated pipes, serving the dual functions of leachate collection and recirculation, as well as gas (CO_2 , CH_4) collection.

(c) Main reaction layer (2–3 m): Materials are placed in alternating succession as "steel slag layer (20 cm) → organic mixture layer (straw, kitchen waste, manure; 15 cm) → steel slag layer → biochar layer (5 cm)," forming 5–8 cyclic units. Water-quenched converter steel slag is preferentially selected. The steel slag is classified by particle size: coarse slag (20–40 mm) for the bottom layer, fine slag (5–15 mm) for the middle layer, and powdered slag (<5 mm) for the top layer, thereby utilizing inter-particle pore size differences to control reaction rate zonation. Black soldier fly larvae are inoculated into the organic mixture layers (initial rate: 5,000 individuals/m², supplemented with 2,000–3,000 individuals/m² after two weeks).

After every 2–3 material layers are placed, moderate compaction is applied using a light vibratory roller (static pressure 0.2–0.5 MPa) to simulate the overburden sediment pressure effect, promoting tight inter-particle contact among steel slag particles while avoiding excessive compaction that would impede gas diffusion.

(d) Surface cover layer (20 cm): Composed of steel slag powder mixed with a small amount of imported topsoil, amended with activated carbon or zeolite (5 wt%) to enhance deodorization. Alkali-tolerant pioneer plant species such as *Suaeda glauca* and *Cynodon*

dactylon, as well as deodorizing plants such as mint (*Mentha* spp.) and lemongrass (*Cymbopogon citratus*), are planted.

(e) Methane oxidation cover layer (30–50 cm): A mixture of compost and wood chips (1:1 by volume), which utilizes naturally enriched methanotrophic bacteria to oxidize CH_4 into CO_2 .

(f) Leachate conditioning tank: Volume approximately 500–800 m^3 (capable of accommodating three months of leachate generation). The tank is divided into a normal recirculation compartment and an overflow temporary storage compartment for the rainy season. A floating cover is installed on top, and the tank is equipped with an online liquid level meter and conductivity monitoring probe. When the conductivity exceeds 8 mS/cm , the corresponding batch of leachate is diverted to an evaporation pond or mixed with steel slag powder to prepare backfill material, and is no longer recirculated.

(g) Black soldier fly breeding area: A mesh-enclosed facility ($\sim 10 \text{ m}^2$) is established adjacent to the heap. Prepupae are periodically collected from the heap and transferred to the breeding area for eclosion, mating, and oviposition. The resulting egg masses are then re-inoculated into the organic mixture layers of the heap.

(h) Monitoring system: Temperature, moisture content, pH, and gas (O_2 , CO_2 , CH_4) sensors equipped with data loggers are embedded at different depths within the heap (50 cm, 100 cm, 200 cm). At least three groundwater monitoring wells are installed around the site (one upgradient, two downgradient), with quarterly sampling and analysis for pH, heavy metals, COD, and other relevant parameters.

4.3 Operation and Regulation Scheme

Months 0–3 (Rapid start-up phase): All layer sequences are placed in a single construction campaign. Black soldier fly and MICP bacterial agents (*Bacillus pasteurii*, *Bacillus sphaericus*, *Bacillus megaterium*, and CA-producing enzyme bacteria) are inoculated. Leachate recirculation is initiated (twice per week, recirculation ratio 0.5:1). Heap temperature (target: 55–65°C) and pH (target: 7–9) are monitored with priority.

Months 4–12 (Main reaction phase): Black soldier flies undergo continuous reproduction and bioconversion. Kitchen waste and straw are supplemented monthly. Leachate recirculation frequency is reduced (once per week). Low-concentration CO₂ flue gas (5–10% CO₂, sourced from plant boiler exhaust) injection into the bottom of the heap begins. Forced aeration is performed once per month (low-pressure blower, airflow rate 0.5–1.0 L/min/kg) to maintain dissolved oxygen concentration within the heap above 2 mg/L. Natural plant colonization commences.

Years 2–5 (Stabilized transformation phase): Supplementation of organic waste is terminated. Leachate recirculation is reduced to once per month. Continuous CO₂ flue gas injection is maintained. Heap sampling (core drilling) is conducted semi-annually for analysis of f-CaO content, CaCO₃ content, mineral phase composition (XRD), micro-morphology (SEM), and pore structure (mercury intrusion porosimetry, MIP). 16S rRNA high-throughput sequencing of the microbial community is performed every three months.

Odor control scheme: Biochar or zeolite (5–10% by volume) is incorporated into the organic mixture layers to increase porosity and promote natural ventilation. Activated carbon or zeolite (5 wt%) is added to the surface cover layer to capture H₂S and NH₃. A biofilter (packed with wood chips, bark, and compost) is installed at the end of the gas collection pipe network to oxidize H₂S, NH₃, and VOCs into SO₄²⁻ and NO₃⁻.

Salt accumulation prevention and control: Approximately 10–15% of the total leachate volume is discharged monthly, and the high-salinity leachate is directed to an evaporation pond or mixed with steel slag powder to produce low-strength backfill material. During the rainy season, fresh water is introduced through a rainwater collection system for dilution. Halophytes (*Suaeda glauca*, *Suaeda salsa*) are planted on the heap surface to absorb salt ions such as Na⁺ and K⁺.

Methane control: When the methane concentration collected from the gas pipe network exceeds 5% (v/v), a ground-level flare system is activated for combustion treatment.

End of Year 5 (Evaluation and mining phase): A comprehensive assessment of the f-CaO conversion rate, mineral phase composition, and physical-mechanical properties of

each heap layer is conducted. The surface zone is retained as an ecological landscape. The lower zones are selectively "mined," and the recovered "artificial ore" is sent to a building materials remanufacturing pilot line for medium-scale testing of autoclaved bricks, carbonated bricks, and aggregates.

4.4 Key Monitoring Indicators

Table 5 Key monitoring indicator system for the 10,000-ton-scale proof-of-concept engineering system

5 Strategic Value and National Significance

5.1 Establishing a National Strategic Reserve of Calcium-Silicon Resources

From a geological resource perspective, the 1.5 billion tons of steel slag historically stockpiled in China, with a CaO content of approximately 40–50% and SiO₂ content of approximately 10–15%, represents a highly substantial quantity of recoverable calcium-silicon resources. Against the backdrop of limited natural limestone resources, increasingly stringent mining restrictions, and the progressive depletion of high-quality sand and gravel aggregates, these steel slag deposits constitute a veritable — and as yet unrecognized — "artificial limestone ore vein." Through a controlled transformation period of 5–10 years, AAD technology converts this "low-grade raw ore" into stable, readily extractable "high-grade artificial ore." In essence, it transforms the current environmental liability into a national strategic resource reserve capable of securing raw material security for the building materials industry for decades to come.

5.2 Generating Massive Artificial Carbon Sinks

The carbonation-based carbon sequestration potential of steel slag is theoretically established: each ton of f-CaO that is fully carbonated can fix 0.78 tons of CO₂. Given that silicate minerals such as C₂S and C₃S also exhibit carbonation reactivity, the comprehensive carbon sequestration potential of steel slag is approximately 0.2–0.4 tCO₂ per ton of steel slag. If the existing 1.5 billion tons of historically stockpiled steel slag

were fully carbonated through the AAD system, 300–600 million tons of CO₂ could be sequestered over the full life cycle. If the approximately 100 million tons of newly generated steel slag each year were also incorporated into this system, an additional 20–40 million tons of CO₂ could be fixed annually. The carbonation product, CaCO₃, represents one of the most stable forms of carbon sequestration on a geological timescale, far surpassing carbon stored in biomass form.

Life-cycle carbon footprint analysis. Carbon sink side (positive benefits) includes: steel slag carbonation, which fixes 0.2–0.4 tCO₂ per ton of steel slag; biochar stable carbon sequestration, which fixes 0.05–0.1 tCO₂ per ton of organic waste; and plant photosynthetic carbon fixation (during the ecological park stage), which sequesters 0.01–0.02 tCO₂ per year per mu (approx. 0.067 ha). Carbon emission side (negative benefits) includes: material transportation; leachate recirculation pumping (annual electricity consumption of approximately 5,000–10,000 kWh); CO₂ injection fans; mechanical compaction; and personnel operations and monitoring. Preliminary estimates indicate a net carbon sink efficiency of approximately 70–80% of the theoretical carbon sequestration capacity, corresponding to a net carbon sink of approximately 0.15–0.3 tCO₂ per ton of steel slag processed.

Carbon sink classification and long-term monitoring. Carbon sinks should be classified into three categories and measured separately: (1) Permanent carbon sink (geological storage grade, half-life >10⁶ years): carbon fixed as CaCO₃ through steel slag carbonation; (2) Long-term carbon sink (centennial scale, half-life approximately 100–1,000 years): carbon fixed in biochar; (3) Temporary carbon sink (annual to decadal scale): residual undecomposed organic carbon and plant biomass carbon remaining in the stockpile. It is recommended to establish a multi-scale carbon sink monitoring system comprising: stockpile-scale monitoring (CO₂ flux sensing probes); landfill-scale monitoring (eddy covariance method for measuring net CO₂ flux); laboratory analysis (TGA + EA for determining total carbon and inorganic carbon content); and isotope tracing (¹³C-labeled CO₂ to track the pathways of carbon migration and transformation).

Temporal value of carbon sinks. Steel slag carbonation achieves immediate carbon sequestration — CO₂ is permanently fixed within hours to days, and the carbon sink

benefit is realized instantaneously. By contrast, afforestation-based carbon sinks require 10–20 years to reach maximum carbon sequestration rates and face reversal risks from wildfires, pests, and diseases. Under the urgent timetable of the "dual carbon" goals (carbon peak and carbon neutrality), technological solutions capable of generating substantial, immediate carbon sinks in the near term possess irreplaceable strategic value. The temporal advantage of steel slag carbonation as an instantaneous carbon sink provides additional support for its pricing and competitiveness in carbon trading markets.

Comparison with other carbon sink technologies. The unique advantages of steel slag carbonation for carbon sequestration include: extremely high carbon stability (CaCO_3 is one of the most stable carbon sink forms in the Earth's surface layer); synergy with solid waste management (simultaneously achieving steel slag valorization and carbon sequestration); and acceptable cost (leveraging natural forces and the leverage of time to reduce energy consumption). In carbon trading markets, immediate carbon sinks typically command a "temporal premium" — buyers are willing to pay higher prices for immediate emission reductions.

Carbon leakage prevention and control. Potential carbon leakage pathways include: carbon emissions arising from land-use changes caused by land occupation for stockpile sites; fuel-related carbon emissions from transporting organic waste; and reduced alternative energy carbon savings when organic waste is diverted from its original treatment pathway (e.g., incineration for power generation). Prevention and control measures include: prioritizing the use of abandoned industrial land and already degraded land for stockpile site construction; prioritizing organic waste sources within a 50 km radius of the stockpile site; and prioritizing organic waste originally destined for landfill disposal to avoid competition with high-energy-efficiency pathways such as incineration for power generation.

Carbon sink measurement and certification. The carbon sink volume from steel slag carbonation has not yet been incorporated into existing carbon trading systems (e.g., CCER methodologies). It is recommended to collaborate with organizations such as the China Quality Certification Center (CQC) to promote the development of a CCER methodology for steel slag carbonation. Prior to approval of the methodology,

certification under international voluntary carbon reduction standards such as VCS (Verified Carbon Standard) or GS (Gold Standard) may be pursued as an interim measure.

5.3 Catalyzing a New Industry: The "Artificial Ore Deposit Industry"

The "Jianggang Plan" is not merely a technology; it has the potential to catalyze an entirely new industrial category — the "artificial ore deposit industry." This industry encompasses the entire chain: the refined geological design of stockpile sites (a form of "reverse mining" that prepares for future extraction); long-term monitoring and process regulation (analogous to mine pressure and hydrological monitoring); and future selective extraction and building material remanufacturing. In terms of industrial form, it represents a hybrid of the "storage + dispatch" model of strategic petroleum reserves and the "ecology + landscape" model of national parks — a composite emerging industry that integrates resource reserve management, carbon sink operations, ecological services, and building materials production.

5.4 Deep Alignment with National Strategies

AAD technology aligns closely with China's top-level national strategies across multiple dimensions: it directly contributes to the carbon sink target of the "dual carbon" goals; it perfectly fulfills the requirements of the "Ten Measures for Solid Waste" concerning "strengthening control over historically stockpiled volumes" and "improving the comprehensive utilization rate of bulk solid wastes"; it responds in a highly innovative manner to the strategic deployment of "building a waste recycling system" in the 15th Five-Year Plan for Circular Economy Development; and the ecological succession outcome of "mining area transformed into park" provides a vivid practical example from the field of industrial solid waste management that embodies the principle that "lucid waters and lush mountains are invaluable assets."

5.5 Preliminary Economic Feasibility Analysis

Revenue source analysis. The main revenue sources include: solid waste treatment fees (CNY 30–50 per ton paid by steel slag producers); organic waste treatment fees

(CNY 100–150 per ton subsidy for food waste treatment, CNY 50–100 per ton subsidy for straw collection); carbon sink revenue (calculated at an expected CCER price of CNY 50–80 per tCO₂); building material product sales (autoclaved bricks/carbonated bricks at CNY 50–80 per ton); and black soldier fly product revenue (insect protein at CNY 3,000–5,000 per ton, insect frass organic fertilizer at CNY 500–800 per ton). At scale, ecotourism revenue may also serve as a supplementary income stream.

Cost analysis. The main costs include: land occupation fees; infrastructure construction costs (HDPE impermeable membranes, pipeline networks, sensors); operation and maintenance costs (energy consumption, labor, monitoring); and interest on investment (five-year capital immobilization cost). Based on preliminary estimates, the comprehensive annual operating cost of a 10,000-ton-scale concept verification project treating 4,000 tons per year is approximately CNY 40–50 million per year. After accounting for subsidies and carbon sink revenues, annual income is estimated at approximately CNY 45–55 million, suggesting the potential to achieve a break-even to marginally profitable operation.

Economies-of-scale analysis. Table 6 presents a comparison of costs and benefits across projects of different scales.

Table 6 Cost–Benefit Comparison of "Jianggang" Projects at Different Scales

As project scale increases, unit costs decline significantly and economic returns progressively improve. It is recommended that, once concept verification is successful, replication and scaling be pursued as expeditiously as possible, first to the 100,000-ton scale and then to the 1-million-ton scale.

Risk assessment and sensitivity analysis. The main risk factors include: technical risk (incomplete diagenesis within five years; probability: medium; impact: high; response: extend the duration to 7–8 years or optimize process parameters); technical risk (decline of functional microbial consortia; probability: medium; impact: medium; response: multiple inoculations + establishment of a backup strain bank); market risk (large fluctuations in carbon sink prices; probability: high; impact: medium; response: sign long-term advance carbon sink sales agreements); market risk (depressed building

materials market; probability: medium; impact: medium; response: product diversification + downgraded utilization); policy risk (adjustments to solid waste subsidy policies; probability: low; impact: high; response: reduce dependence on any single subsidy); and environmental risk (heavy metal leakage; probability: low; impact: high; response: multi-layer impermeable barriers + continuous monitoring).

Sensitivity analysis indicates that, using a carbon sink price of CNY 60 per tCO₂ and a steel slag treatment fee of CNY 40 per ton as the baseline: when the carbon sink price falls by 50% (to CNY 30 per tCO₂), project revenue decreases by approximately 15–20%; when the steel slag treatment fee falls by 50% (to CNY 20 per ton), project revenue decreases by approximately 10–15%; when both the carbon sink price and the steel slag treatment fee fall by 50% simultaneously, the project can still maintain near-break-even operation. This indicates that the project's economic performance exhibits good resilience to variations in key parameters.

5.6 Exploration of Innovative Business Models

In addition to the traditional models of solid waste treatment fees and building material sales revenue, the following innovative business models are recommended for exploration:

"Carbon Sink Bank" model: At project inception, steel enterprises sign forward carbon sink pre-sale agreements with carbon sink demanders, locking in carbon sink prices in advance (analogous to carbon futures), thereby securing upfront investment capital.

"Green Trust" model: The project is packaged as a green trust product and offered to institutional and individual investors for fundraising, with the project's diversified revenue streams serving as the repayment source.

"Government Service Procurement" model: The project is incorporated into the government's public service system for ecological restoration and solid waste management. Through franchising or service procurement arrangements, the government pays the project operator ecological restoration service fees and carbon sink procurement

fees. This model has established precedent in public infrastructure sectors such as wastewater treatment, waste incineration, and mine site restoration. The government can provide stable support by: including the project in local special plans for solid waste treatment facilities (granting land allocation and tax incentives); signing long-term (10–20 year) solid waste treatment service contracts (securing the project's baseline revenue); and incorporating the carbon sink into local carbon peak action plans.

"Ecological Restoration + Resource Development" integrated model: Stockpile sites are combined with mine ecological restoration and urban brownfield remediation projects. The local government provides land and supporting infrastructure, while the operator is responsible for construction and operation, thereby achieving the unification of ecological benefits and economic returns.

6 Risks, Challenges, and Further Research Needs

6.1 Key Scientific Issues

(a) Multi-factor synergistic kinetic modeling. The reactions occurring simultaneously within the stockpile—including organic acid leaching, microbial mineralization, direct carbonation, and hydration—exhibit complex coupling and competitive relationships. Establishing a mathematical model that accurately describes the transformation kinetics of f-CaO and the RO phase (MgO-FeO-MnO solid solution) under multi-factor synergy is a prerequisite for process prediction and parameter optimization. Currently lacking are experimental data on steel slag transformation kinetics under the ternary coupling condition of "organic acid–MICP–CO₂."

(b) Multi-field coupling of mass transfer, heat transfer, and reaction within the stockpile. A 10,000-ton-scale stockpile constitutes a典型的 heterogeneous porous media reactor, within which the distribution and evolution of the temperature field, moisture field, gas concentration field, and chemical reactions are extremely complex. Multi-field coupled numerical simulation methods tailored to steel slag–organic waste mixed stockpiles need to be developed.

(c) Long-term maintenance of functional microbial activity. Whether the inoculated MICP functional bacterial consortium can maintain its activity over a 5-year timescale within the continuously evolving chemical and physical environment of the stockpile is the core microbiological issue determining system success. Long-term experiments are needed to study the succession patterns and functional stability of the microbial community. Bacteriophage infection is a significant risk factor that can lead to functional microbial collapse; preventive and control measures include multi-strain rotational inoculation, physical isolation via biochar micropores, environmental regulation deviating from optimal phage reproduction conditions, and the establishment of a backup strain bank.

(d) Long-term stability of the RO phase. The RO phase (MgO-FeO-MnO solid solution) undergoes slow hydration under natural conditions, which may still cause volume changes over a timescale of decades. In the AAD environment, the organic acid environment of Phase I can selectively react with MgO on the surface of the RO phase, converting it into dissolved Mg^{2+} , which is subsequently carbonated into stable MgCO_3 . The transformation effectiveness of the RO phase should be evaluated by monitoring the amount of high-magnesium calcite formed.

(e) Heavy metal migration, transformation, and risk control. Heavy metals in steel slag (Cr, Ni, Pb, Zn, Cd, etc.) face the risk of activation and leaching under the low-pH conditions (pH 4–6) of Phase I. However, the AAD system possesses a unique "self-stabilization" mechanism: when the pH rises back to 8–10 in Phases II and III, dissolved heavy metals can be re-immobilized through co-precipitation with CaCO_3 , adsorption by C-S-H gel, and complexation with oxygen-containing functional groups on biochar surfaces. Recommendations: control organic acid production during the acidification phase to avoid prolonged pH below 5; increase biochar dosage (from 5% to 8–10%); conduct monthly sampling to monitor key heavy metal concentrations (Cr(VI), Ni, Pb, etc.) in leachate; and establish an emergency leachate collection pond.

(f) Steel slag–black soldier fly compatibility. The dose–response relationship between steel slag incorporation ratio and black soldier fly growth and development must be determined experimentally. Experimental design: set up steel slag incorporation

gradients (0%, 10%, 20%, 30%, 40%, 50% by mass) under laboratory conditions, introduce 100 three-day-old larvae per group, observe continuously for 14 days, and monitor larval survival rate, average body weight, pupation rate, adult emergence rate, and heavy metal content in larvae and frass. Simultaneously, set up frass–steel slag transformation experiments (frass:steel slag mass ratio gradient), cultivate under constant temperature and humidity for 90 days, and monitor changes in f-CaO and CaCO_3 contents. It is expected that the "slow-release" characteristics of humic acid and fulvic acid in the frass will continue to exert a promoting effect during Phases II and III.

6.2 Key Engineering Challenges

(a) Leachate management and recirculation optimization. The leachate generated by the stockpile contains high concentrations of calcium ions, organic acids, and microbial metabolites. The timing, flow rate, and water quality control of its recirculation require fine-tuned regulation. When the COD in the leachate exceeds 5,000 mg/L or the $\text{NH}_3\text{-N}$ concentration exceeds 500 mg/L, a pretreatment pond should be introduced to utilize the inherently high pH and high calcium characteristics of steel slag for chemical precipitation pretreatment before recirculation.

(b) Product homogeneity assurance. Due to spatial heterogeneity in temperature, moisture, gas concentration, and microbial distribution within the stockpile, the mineral composition and mechanical properties of the "artificial ore" may vary significantly across different zones. Assurance measures include: rationally designing the spatial distribution of leachate recirculation and CO_2 injection points; adopting zonal extraction combined with mechanical stirring homogenization; and establishing a graded utilization system—high-activity products ($\text{CaCO}_3 > 60\%$, f-CaO $< 1\%$) for cement clinker, medium-activity products (CaCO_3 30–60%) for autoclaved products, and low-activity products ($\text{CaCO}_3 < 30\%$) for carbonated building materials or engineering fillers.

(c) Ecological risk prevention and control. The potential impact of the system on the surrounding environment should be assessed from the following dimensions: groundwater safety (upstream and downstream monitoring wells, quarterly testing for pH, heavy metals, and COD, with early warning thresholds set); atmospheric environmental

impact (monitoring H₂S at the stockpile boundary not exceeding 0.06 mg/m³); soil and biodiversity (semiannual heavy metal testing of soil within 50 m of the stockpile perimeter, and surveys of plant and soil animal diversity).

(d) Winter low-temperature countermeasures. In northern regions, the optimal temperature for black soldier fly (25–30°C) is difficult to maintain naturally. Countermeasure strategies: lay a 30–50 cm thick insulation layer of straw or wood chips around the bottom and sides of the stockpile, utilizing internal bioheat to maintain core temperature; breed low-temperature-adapted black soldier fly strains; and adopt a rotational scheme of "black soldier fly as the primary transformation agent from April to October + microorganism-dominated phase from November to March."

6.3 Further Research Roadmap

Table 7 "Jianggang Plan" Phased Research Roadmap

7 Conclusions

The "Jianggang Plan" proposed in this paper represents a fundamental leap in the perception of "industrial solid waste." It acknowledges the temporary limitations of human engineering interventions within the grand material cycle and instead turns to nature for solutions, harnessing its most powerful weapons—time and biological communities—to resolve the century-old challenge of steel slag resource utilization.

The core academic contributions of this paper are as follows:

For the first time, the theory of "Accelerated Artificial Diagenesis (AAD)" has been systematically proposed, compressing geological-scale natural diagenesis into an engineering-achievable 5-year cycle through multi-factor synergistic intensification;

A multi-source waste synergistic共生体系 has been constructed, revealing five key biochemical mechanisms: the coupling of organic acid release with calcium ion release, the active utilization of steel slag self-pulverization, the coupling of bioheat generation with self-heating maintenance, the coupling of in-situ CO₂ supply with carbonation, and the coupling of straw skeleton structure with enhanced aeration;

A "microorganism–insect–plant" tripartite synergistic transformation system has been designed, achieving the organic integration of functional bacterial mineralization, black soldier fly bioconversion, and pioneer plant ecological restoration;

A five-year, three-phase ecological succession model has been proposed, establishing a rigorous correspondence with geological diagenetic stages;

A complete conceptual engineering design at the 10,000-ton scale has been completed, demonstrating its scientific feasibility, engineering viability, and economic rationality.

This is not merely a technical solution, but a new resource perspective and development philosophy. Its core insight is that the industrial waste currently regarded as a burden, under appropriate spatiotemporal scales and ecosystem design, can fully be transformed into a strategic resource for the future. This paradigm shift—from "treatment" to "transformation" and from "burden" to "asset"—may well represent the starting point for industrial civilization to move toward a "geological-scale circular economy" in harmony with the Earth system.

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